EXAM REVISION 1

MATHEMATICS PAPER 1 – GRADE 10

- This revision paper comprises 21 pages with 9 questions, with many sub-questions.
- As many key concepts have been covered as possible.
- Grade 11 concepts sometimes covered in Grade 10 are not included.
- Answer all questions if possible.
- Guideline marks are provided for each question.
- Do not be concerned with the high total number of marks in some questions/sub-questions. This just facilitates more mathematical concepts being included for practise.
- Number your answers exactly as described in the questions.
- Clearly show all workings and calculations, as applicable.
- Correct answers only will not necessarily generate full marks.
- A non-programmable calculator may be used.
- All answers should be rounded to TWO decimal places unless otherwise directed.
- Diagrams are not necessarily drawn to scale.

NOTE: A Memorandum may be provided on request. It is certainly in your interest to attempt and apply your mind to all questions prior to receiving or making use of the memorandum Use self-marking to reinforce understanding of all concepts covered.

Think twice, write once, but most of all, ENJOY!

All the very best in all your school exams!

NB: Every effort has been made to present accurate questions and solutions, however please note that no third party checking has been applied whatsoever.

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NUMBER SYSTEM, ALGEBRA, EQUATIONS, INEQUALITIES, EXPONENTS

QUESTION 1

1.1 Given the set: $\left\{\sqrt{4}\ ;\ \frac{1}{3}\ ;\ \sqrt{7}\ ;0\ ;\ \sqrt{\frac{25}{9}}\ ;\ \sqrt{-16}\ ;\ \sqrt[3]{-27}\ ;0,\dot{2}\ ;\ \pi\right\}$, list all the rational numbers in the set. (4)

1.2

1.2.1 How many rational numbers are there between $\frac{4}{5}$ and $\frac{17}{20}$? (1)

1.2.2 Showing all working, write down a rational number between $\frac{4}{5}$ and $\frac{17}{20}$. (3)

1.2.3 Prove that the difference of two rational numbers will always be a rational number. (Hint: Let the numbers be a/b and c/d, with $b, d \neq 0$) (4)

1.3 Answer True of False:

1.3.1 The difference between two natural numbers will always be a natural number. (1)

1.3.2 The division of two whole numbers will always be a rational number. (1)

1.3.3 The sum of two integers will always be an integer. (1)

1.4 Given that $T = \sqrt{\frac{8}{1-x}}$, write down a value of x for which T will be:

1.4.1 Undefined (1) 1.4.2 Non-Real (1) 1.4.3 An integer (1)

1.5 Without the use of a calculator, and showing all working, express each of the following as a common fraction in its <u>simplest</u> form.

1.5.1 $0,1\dot{2}\dot{3}$ (4) 1.5.2 $0,2\dot{3}$ (4)

1.6 Between which two consecutive integers does $-2\sqrt{7}$ lie? (3)

1.7 <u>Without the use of a calculator</u>, and showing <u>all</u> working, show that:

$$\sqrt[3]{25} > 2\sqrt{2} > \sqrt[6]{500} \quad (3)$$

1.8 If $P = 3\left(\frac{x}{y} - 2\right)$, determine the value of P if:

1.8.1 $x = 6$ and $y = 2$ (2) 1.8.2 $x = 0$ and $y \neq 0$ (2) 1.8.3 $x = y$ and $y \neq 0$ (2)

1.9 Prove that the product of two different odd numbers is always odd. (3)

(Hint: Let the numbers be $2n + 1$ and $2m + 1$)

1.10 Make V the subject of the following equation: $W = \frac{1}{M} - \frac{2}{V}$ (3)

1.11 Make y the subject of the following equation: $\frac{1}{x} = \frac{1}{y} + \frac{1}{w}$ with $x, y, w \neq 0$ (3)

1.12.1 Make T the subject of the following equation: $P = \sqrt{\frac{1+T}{1-T}}$, $T \neq 1$ (3)

1.12.2 Noting the restriction on the value of T above, use your answer to prove that T cannot equal 1. (2)

1.13 The following formula converts degrees Fahrenheit to degrees Celsius:

$$°C = \frac{5}{9}(°F - 32)$$

1.13.1 Use this formula to derive a formula which may be used to convert degrees Celsius to degrees Fahrenheit. (2)

1.13.2 Establish the single temperature where, irrespective of whether one uses a Fahrenheit or a Celsius thermometer, the temperature read will be the same. (2)

1.14 Given $A = \frac{\sqrt{x+3}}{4}$, write down ONE value of x which will make A:

1.14.1 A whole number (1) 1.14.2 Equal to zero (1)

1.14.3 Non-real (1) 1.14.4 Irrational (1)

QUESTION 2

2.1 Fully simplify the following:

2.1.1 $(m + 2n)(3m - 2n) + 4n^2$ (3) 2.1.2 $\frac{1}{4}(2x - 6)^2$ (3)

2.1.3 $\frac{12x^4y}{4xy^3} \div \frac{3x^2}{y^2} - (x + y)^0$ (4) 2.1.4 $\frac{8x+3}{4} - \frac{2x-1}{2} - \frac{5}{4}$ (4)

2.1.5 $\frac{x^2-9}{4x+12} \times \frac{6x^3-2x^2}{6x^2-20x+6} \times 4$ (6) 2.1.6 $\frac{x}{x-1} - \frac{2}{x+2} + \frac{6}{-x^2-x+2}$ (6)

2.1.7 $\left(a^{\frac{1}{3}} \times b^{\frac{2}{3}} \times c^{\frac{1}{2}}\right)\left(a^{\frac{2}{3}} \times b^{-\frac{2}{3}} \times c^{\frac{1}{2}}\right)$ (3) 2.1.8 $\left(x + \frac{1}{x}\right)\left(\frac{1}{x} + x\right)$ (3)

2.1.9 $\frac{5x^6 \,.\, y^{-2}}{15 \,.\, x^4 \,.\, y^4} \div \frac{1}{3}$ (3) 2.1.10 $x^{-1} + y^{-1} \div \left(\frac{x+y}{x^2y^2}\right)$ (4)

2.1.11 $\frac{2x^2+10x+12}{\frac{1}{2}(x+3)}$ (4)

2.1.12 $\frac{x^2-4+ax+2a}{2-x-a}$ (5)

2.1.13 $\frac{2x+10}{x^3+125} \times \frac{x^2-5x+25}{2}$ (4)

2.1.14 $\frac{5n-5}{5n} \div \frac{n^3-1}{n^2+n+1}$ (5)

2.1.15 $\frac{1}{x}+\frac{3}{2}-\frac{1}{3x}$ (2)

2.1.16 $\frac{4}{m+2}-4+\frac{10m-2}{2m+4}$ (5)

2.1.17 Write down the restrictions on x in 2.1.6 above. (2)

2.1.18 Write down the restriction on x in 2.1.11 above. (1)

2.2

2.2.1 If $(x-y)^2 = 20$ and $x^2+y^2 = 10$, express y in terms of x. (4)

2.2.2 Hence write down any restriction in the value of x, and in the value of y. (2)

QUESTION 3

3.1 Fully factorise the following:

3.1.1 $3a^4 - 48$ (3)

3.1.2 $x^2 - y + x^2y - 1$ (4)

3.1.3 $2(a+b)^2 - a - b$ (3)

3.1.4 $3x^3 + 24y^3$ (3)

3.1.5 $x^5 - x^4 + 1 - x$ (5)

3.1.6 $11a^2 - 22a - 33$ (3)

3.1.7 $x^3 - 4x$ (3)

3.1.8 $2x^2 - 2x - 24$ (3)

3.1.9 $9m^2 - 18m + 2n - mn$ (4)

3.1.10 $(x-2)^2 - 9$ (3)

3.1.11 $\frac{1}{5}x^3 + 25$ (4)

3.1.12 $\frac{1}{2}x^2 - \frac{9}{2}$ (3)

3.2 Given $P = \sqrt{9n^2 - 24n + 16}$

3.2.1 Simplify fully. (3)

3.2.2 Hence determine the value of n for which P will be a minimum. (2)

3.3 Rectangle $ABCD$ has an area of $8m^3 - 27$ units^2, with length AB and width BC of integer size.

3.3.1 Determine the length AB in terms of m if $BC = 2m - 3$ units. (4)

3.3.2 Determine the minimum value of m. (2)

3.3.3 Hence calculate the minimum area of $ABCD$. (1)

QUESTION 4

4.1 Given the inequality $-3 < 1 - 2m \leq 11$, where $m \in \mathbb{R}$:

4.1.1 Solve for m , and express your answer in interval notation. (4)

4.1.2 Represent your answer on the real number line. (2)

4.2 Show the following on a number line: $\{x: x < -1 ; x \in \mathbb{R}\} \cup x \in [2 ; 6)$ (3)

4.3 Given that x must satisfy $-16 \leq 2x < 8$ AND $-x + 1 \leq 4$:

4.3.1 Solve for x in EACH case and represent your answers one above another on the SAME number line. (5)

4.3.2 Hence solve for x satisfying BOTH inequalities, giving your answer in interval notation. (2)

4.4 Given the function $f(x) = x^2 - x - 12$, solve for x if:

4.4.1 $f(x) = 0$ (3) 4.4.2 $f(x) \geq x^2 + 2x$ (4)

4.5 Solve for x:

4.5.1 $-2x^2 - 5x + 12 = 0$ (3) 4.5.2 $2x^2 = 3(5x - 6)$ (3)

4.5.3 $\frac{x-1}{x+2} = \frac{x+2}{3x}$ (4) 4.5.4 $\frac{12}{x-3} + \frac{5}{x} = -\frac{2}{3x-x^2}$ (5)

4.5.5 $3^x + 3^{x+2} = 810$ (4) 4.5.6 $5.8^{x+1} = 320$ (4)

4.5.7 $27^x = \frac{9}{3^{x-1}}$ (4) 4.5.8 $(2^{2x-1})^2 = 64$ (4)

4.5.9 $\frac{2x+1}{3} - \frac{x}{2} = -\frac{x}{6} + 1$ (3) 4.5.10 $(x^2 - m)(x + n) = 0\,,\ m \geq 0$ (3)

4.5.11 $(2x + 1)^2 = (x - 1)(x + 1) + 1$ (4) 4.5.12 $x - \frac{20}{x} = 1$ (4)

4.5.13 $\frac{(5^x)^3 + 5^{3x-1}}{5^{2x}} = 150$ (5) 4.5.14 $\frac{9^x - 1}{3^x - 1} = 82$ (4)

4.5.15 $x^3 + 4x^2 - x - 4 = 0$ (4)

4.6 Solve simultaneously for x and y:

4.6.1 $2x - y = 1$ and $-2 + 3y = 5x$ (4) 4.6.2 $x + y = 15$ and $2^x = 4^y$ (4)

4.6.3 $x + 2y = 7$ and $5x - 3y + 4 = 0$ (4)

4.6.4 $\frac{x}{y} = \frac{2}{3}$ and $y + \frac{1}{2}x = 4$ (3) 4.6.5 $4^x . 2^y = 1$ and $\frac{7^{y+3}}{49^x} = \frac{1}{7}$ (6)

4.6.6 $\frac{x-1}{3} = y + 2$ and $2x + 3 = 8 - 3y$ (4)

4.6.7 $\frac{3}{x}+\frac{6}{y}=2$ and $\frac{1}{x}=\frac{1}{y}-\frac{5}{6}$ (5)

4.7 How many integer solutions are there to the following inequality if $m<10$?

$$\frac{m-2}{4}+1 > \frac{1-m}{3}+2 \quad (5)$$

4.8 Given the inequality $mx-nx-5m+5n>0$, solve for x if:

4.8.1 $m>n$ (2) 4.8.2 $m<n$ (2)

4.9 Determine all possible ratios $\frac{x}{y}$ for which: $2x^2+5xy-3y^2=0$ (4)

4.10 Simplify fully: $\frac{10^{2024}-10^{2022}}{100^{1012}+10^{2023}}$ (4)

4.11 A farmer stores tools in a small rectangular toolshed measuring 4×3 metres. The farmer wishes to increase the size of the toolshed to $35{,}75m^2$ by increasing both the length and the width of the existing toolshed by x metres.

Algebraically, determine the value of x. (5)

4.12 Ten years ago, Shamla was half Benji's age. In twenty years she will be $\frac{6}{7}^{th}$ Benji's age. How old are Shamla and Benji now? (Hint: Let Shamla's age $= x$ and Benji's age $= y$) (6)

4.13 Consider the fraction $\frac{x}{y}$ in its simplest form. If one is added to the numerator, the fraction reduces to $\frac{1}{2}$. If three is added to the denominator, the fraction reduces to $\frac{1}{3}$. Determine the fraction $\frac{x}{y}$ in its simplest form. (6)

4.14 The sum of two numbers x and y is 12. The difference between the larger number and five times the smaller number is 30. Determine x and y. (6)

4.15 ab is a 2-digit number with digits a and b. The digits are reversed to give the 2-digit number ba, with $ba-ab=36$. Determine the value of $(b-a)$. (4)

4.16 In 2024 Brian is twice as old as Chelsea and five years younger than Alan. Their average age is 20. How old will Chelsea be in 2029? (6)

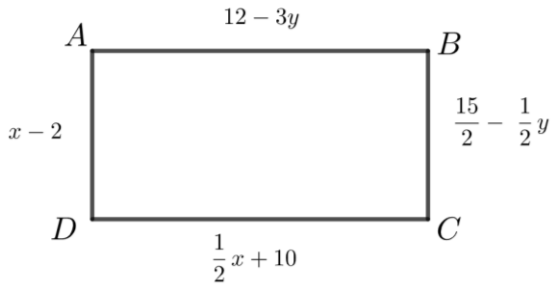
4.17 Alan travels from his home to a dam at an average speed of 75km/h. Later there is a lot more traffic on the same road and he returns home at an average speed of 60km/h. If he takes 45 minutes longer on his return trip, determine:

4.17.1 The distance between Alan's home and the dam. (4)

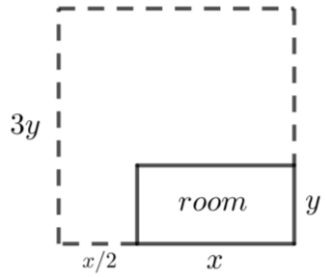
4.17.2 His average speed for the total trip. (3)

(Note: Distance = Speed × Time)

4.18 In the figure, $ABCD$ is a rectangle. Solve simultaneously for x and y. (6)



4.19 A rectangular room has a length x metres and a width y metres. The room in enlarged by adding $\frac{1}{2}x$ to its length and $2y$ to its width, as shown.



The perimeter of the room changed from 14m to 30m.

Determine the values of x and y. (5)

4.20 Sue buys x eggs for R720 and puts 10% of them into the fridge for her own use. She sells the balance of the eggs, each for R1 more than she paid for each of them, and makes a total profit of R144.

4.20.1 Write down the cost of each egg in terms of x. (1)

4.20.2 Write down the number of eggs she sells in terms of x. (2)

4.20.3 Write down the selling price of each egg in terms of x. (1)

4.20.4 Hence determine how many eggs Sue bought. (4)

QUESTION 5

5.1 Simplify the following expressions without the use of a calculator:

5.1.1 $\frac{9^2 \times 3^{-3}}{3^{-1}}$ (2)

5.1.2 $\frac{(4\times3)^3}{3^2\times2^2} - \left(\frac{5\pi}{7^2}\right)^0$ (3)

5.1.3 $\frac{12^{n+1} \cdot 9^{2n-1}}{36^n \cdot 8^{1-n}}$ (5)

5.1.4 $\frac{5^{n-1}+5^{n+1}+5^n}{5^n+5^{n+2}+5^{n+1}}$ (4)

5.1.5 $\frac{4^{n-1} \cdot 2^{2n+3} \cdot 2^2}{2^n \cdot 8^n}$ (5)

5.1.6 $\frac{5^{x+1}+7 \cdot 5^x}{-(5^x-5^{x+1})}$ (4)

5.1.7 (i) $\frac{2^{-n} \cdot 2^{n-2} \cdot 4^{n+1} \cdot 3^{2n+1}}{9^n \cdot 16^n}$ (5)

5.1.7 (ii) Hence or otherwise solve for x: $\frac{2^{-x} \cdot 2^{x-2} \cdot 4^{x+1} \cdot 3^{2x+1}}{9^x \cdot 16^x} = 48$ (3)

5.1.8 $\frac{\left(\frac{x}{3}\right)^{-1}-\left(\frac{y}{2}\right)^{-1}}{\frac{9}{x^2}-\frac{4}{y^2}}$ (5)

5.1.9 Given the following expression: $\frac{81^x \cdot 3^{2-x}-27^x}{8 \cdot 3^x \cdot 3^x \cdot 3^x}$

5.1.9.1 Simplify the expression. (5) 5.1.9.2 Hence complete the following:

"The given expression is _____ of the value of x" (1)

5.1.9.3 Use $x = 0$ as an example to verify your answer in 5.1.9.1 above. (2)

5.1.10 Prove that the following expression is independent of the value of x provided that $x \neq -\frac{1}{3}\,;1\,;\frac{3}{2}$ or -2:

$$\frac{x^2+x-2}{3x^2-2x-1} \div \frac{2x^2+x-6}{9x^2+6x+1} \times \frac{8x-12}{3x+1} \quad (6)$$

NUMBER PATTERNS

QUESTION 6

6.1 The following gives the first four terms of an arithmetic sequence:

$$y\,;2x+4\,;x+4\,;1-x$$

6.1.1 Determine the values of x and y. (5)

6.1.2 Hence determine T_n. (2)

6.1.3 Which is the first 3-digit term? (2)

6.1.4 Prove that 1025 is not a term in the sequence. (2)

6.1.5 What is the value of $T_{39}+T_{40}+T_{41}$. (3)

6.1.6 All odd valued terms are removed from the sequence. Determine the 40th term of the remaining sequence. (5)

6.2 Consider the following number pattern: 3 ; 2 ; 7 ; 2 ; 11 ; 2 ; 15 ; 2 ; 19 ; … … … ..

6.2.1 Write down the value of the 1000th term. (1)

6.2.2 Determine the value of $T_{501} - T_{500}$. (5)

6.2.3 All even valued terms are removed from the pattern. Which is the greatest 3-digit term in the remaining sequence? (4)

6.3 Consider the following number pattern: $\frac{79}{2}$; $\frac{76}{5}$; $\frac{73}{8}$; $\frac{70}{11}$; … … …

6.3.1 Write down the next two terms if the pattern continues in the same manner. (2)

6.3.2 Determine the general term T_n. (4)

6.3.3 How many positive terms are there in the pattern? (3)

6.3.4 Determine the value of n if $T_n = -\frac{8}{89}$ (3)

6.3.5.1 Determine the value of n for which T_n will be as close to 1 as possible. (3)

6.3.5.2 Hence calculate the value of this term, leaving your answer as a fraction. (2)

6.4 An arithmetic sequence is given as follows: -7 ; -3 ; x ; 5 ; 9 ; y ; 17 ; … … …

6.4.1 Write down the values of x and y. (2)

6.4.2 Determine the general term $T_n = bn + c$. (3)

6.4.3 What is the value of the 100th term? (1)

6.4.4 Which term has a value of 869? (2)

6.4.5 Which value of n yields the first 4-digit term? (3)

6.5 Consider the number pattern: -25 ; -22 ; -19 ; x ; -13 ; … … …

6.5.1 Write down the value of x. (1)

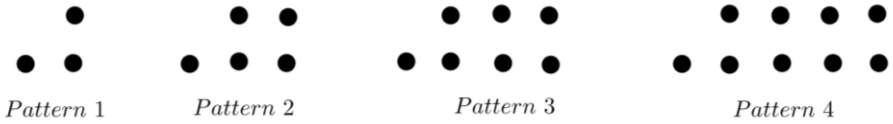
6.5.2 Determine the general term. (3)

6.5.3 Which term equals 92? (2)

6.5.4 Determine the value of the first term greater than 500. (3)

6.5.5 Explain why no term in the pattern is divisible by 3. (2)

6.6 Consider the following sequence of dots:



6.6.1 How many dots are in the 5th pattern? (1)

6.6.2 Determine the general term T_n which defines this pattern. (3)

6.6.3 Which pattern number is the first to contain over 550 dots? (2)

6.6.4 Does a pattern with 6046 dots forms part of this sequence. Explain. (2)

6.7 Consider the following number pattern: $250\ ;\ 3\ ;\ 246\ ;\ 3\ ;\ 242\ ;\ 3\ ;\ 238\ ;\ \ldots\ \ldots\ ;\ -146$
If the pattern continues in the same manner:

6.7.1 Write down the value of the 72nd term. (1)

6.7.2 What is the value of the 101st term? (5)

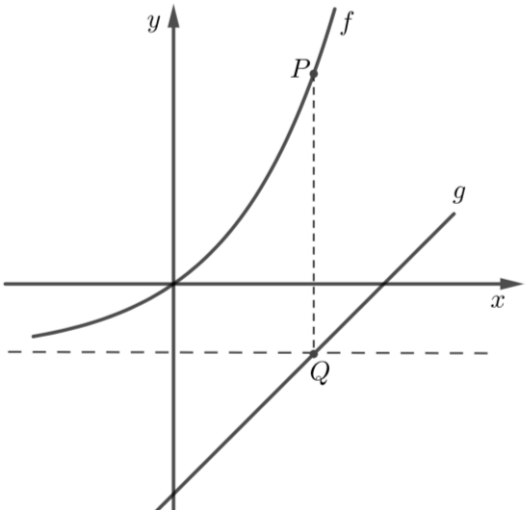
6.7.3 How many terms are there in total? (4)

6.7.4 What is the value of the largest negative term? (4)

FUNCTIONS

QUESTION 7

7.1 $f(x) = b^x + d$ and $g(x) = x - 3$ are shown in the diagram. g intersects the asymptote of f at Q. PQ is a line parallel to the y-axis with P on f and $PQ = 4$ units.



7.1.1 If the range of $f(x)$ is $y > -1$, write down the value of d. (1)

7.1.2 Hence determine the coordinates of Q. (2)

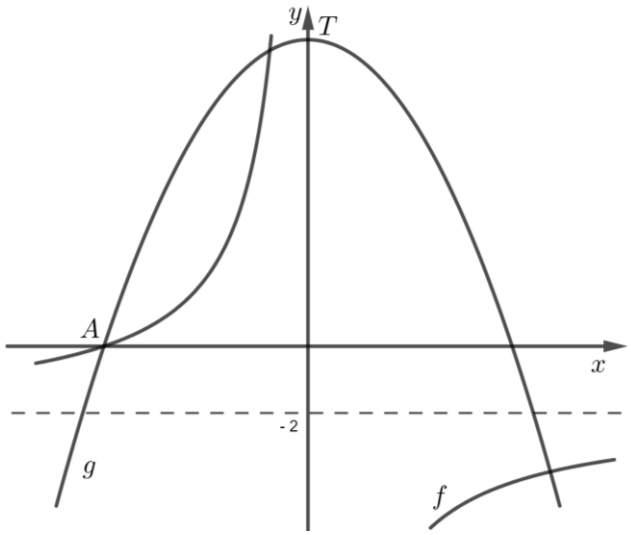
7.1.3 Hence determine the equation $f(x)$. (4)

7.1.4 Prove that f passes through the origin. (2)

7.1.5 Hence determine the distance between the y-intercepts of f and g. (2)

7.1.6 Using interval notation, for which values of x is $f(x) < 0$? (1)

7.2 The graphs of $f(x) = \frac{m}{x} + n$ and $g(x) = -x^2 + 9$ are shown. T is the turning point of g. f and g share a x-intercept at A. A horizontal asymptote $y = -2$ is shown.



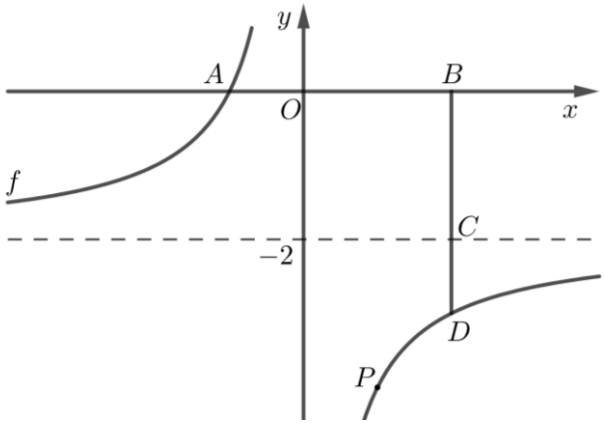
7.2.1 Write down the value of n. (1)

7.2.2 Determine the equation of $f(x)$. (4)

7.2.3 The graph of g is reflected about the x-axis. Write down the coordinates of T^1, being the transformed point T. (1)

7.2.4 Calculate the coordinates of the intersections of f with the line $y = -2x - 6$. (5)

7.3 The graph of $f(x) = \frac{a}{x} + q$ with x-intercept A is shown. $P(1\ ;\ -4)$ lies on f. $BCD \parallel y$-axis with D on f and C on the asymptote of f.



7.3.1 Determine the equation $f(x)$. (3)

7.3.2 Calculate the length OA. (3)

7.3.3 Determine length CD if $OB = 2$ units. (3)

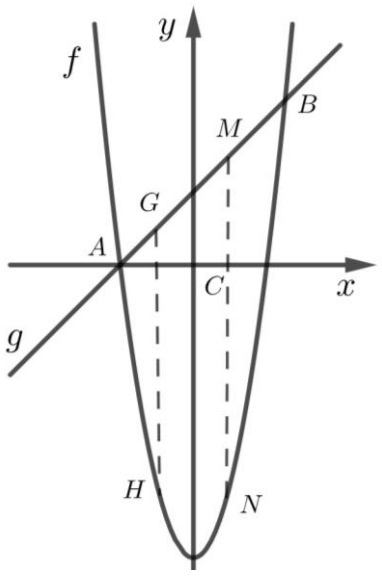
7.3.4 Write down the equation of the axis of symmetry with a negative gradient. (1)

7.3.5 Hence determine the x-coordinate of M and N, being the points of intersection between the equation in 7.3.4 and f, with M in the 4th quadrant. Give your answers in surd form. (4)

7.3.6 Write down the range of $f(x)$. (1)

7.3.7 Determine the value(s) of x for which the perpendicular distance from f to the x-axis is 3 units. (4)

7.4 The graphs of $f(x) = 2x^2 + q$ and $g(x) = x + 2$ are shown. The graphs share a common x-intercept at A and they intersect once more at B.



7.4.1 Determine the value of q. (3)

7.4.2 If $f(x) = 2x^2 - 8$, determine the coordinates of B. (4)

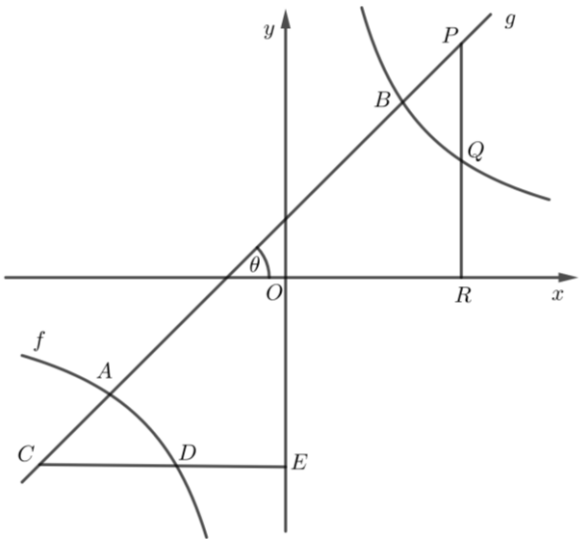
7.4.3 For which values of x is $f(x) > g(x)$? (2)

7.4.4 For which values of x is $f(x) \leq 0$? (2)

7.4.5 The line $GH \parallel y$-axis has G on g and H on f. GH intersects the x-axis at -1. Determine the length GH. (3)

7.4.6 The line $MCN \parallel y$-axis has M on g and N on f such that MN can move between A and B. C is the x-intercept of MN. Determine the length(s) AC if $MN = 10$ units. (5)

7.5 In the sketch the graphs of $f(x) = \frac{m}{x}$ and $g(x) = nx + 1$ are shown. The graphs intersect at A and $B(2\,;3)$. $PQR \perp x$-axis with P on g and Q on f. $CDE \perp y$-axis with C on g and D on f.



7.5.1 Determine the value of m. (2)

7.5.2 If $\theta = 45^\circ$, write down the value of n. (1)

7.5.3 Hence determine the coordinates of A. (3)

7.5.4 If $PQ = 2$ units, determine:

7.5.4.1 Length OR. (4) 7.5.4.2 The coordinates of P and Q. (3)

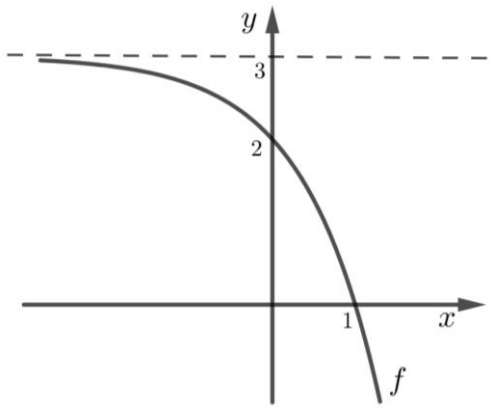
7.5.5 Determine length CD if $OE = 4$ units. (3)

7.5.6 Write down the equation of the axis of symmetry of $f(x)$ with a +ve gradient. (1)

7.5.7 Explain why the line in 7.5.6 will never intersect g. (1)

7.5.8 $h(x) = f(x) + k$. Determine the value of k if the x-intercept of h is $\frac{1}{2}$. (4)

7.6 In the sketch $f(x) = am^x + n$ is shown. The equation of the horizontal asymptote is $y = 3$.



7.6.1 Write down the value of n. (1)

7.6.2 Hence determine the equation of $f(x)$. (4)

7.6.3 For which values of x is $f(x) < 0$? (1)

7.6.4 Write down the range of $f(x)$. (1)

7.6.5 $g(x) = f(x) - \frac{8}{3}$. Determine the x-intercept of g. (3)

7.6.6 $h(x) = f(x) + k$. For which value(s) of k will h not intersect the x-axis? (2)

7.7

7.7.1 Sketch the graphs of $f(x) = \frac{-2}{x} - 2$ and $g(x) = -\frac{1}{2}x - 2$ on the same set of axes. Clearly show all applicable axes intercepts, the asymptote, and at least one useful plotting point for f. Label the points of intersection of f and g as A and B with B in the 4th quadrant. (7)

7.7.2 Determine the coordinates of A and B. (4)

7.7.3 For which value(s) of x is:

7.7.3.1 $f(x) > 0$? (1) 7.7.3.2 $g(x) \leq f(x)$? (2) 7.7.3.3 $\frac{g(x)}{f(x)} = 0$? (1)

7.8 $f(x) = a^x + q$ and $g(x) = 2x + k$. The range of f is $y \in (-3\,;\,\infty)$ and $P(-2\,;6)$ lies on f. f and g share a common y-intercept.

7.8.1 Determine the equation of $f(x)$. (4)

7.8.2 Hence determine the value of k. (2)

7.8.3 Plot the two graphs clearly showing all applicable axes intercepts and asymptote. Label the x-intercepts of f and g as A and C respectively. (5)

7.8.4 Write down the length AC. (1)

7.8.5 For which values of x are both $f(x)$ and $g(x)$ simultaneously < 0 ? (2)

7.9

7.9.1 Neatly sketch the graph of $f(x) = 2x + 4$ in the domain $x \in [-3\,;1]$. Clearly show all axes intercepts. (4)

7.9.2 Write down the gradient of f. (1)

7.9.3 If the graph of $g(x) = kx + 2$ is perpendicular to f, write down the value of k. (1)

7.9.4 What is the range of f? (2)

7.9.5 The graph of $h(x) = ax^2 + c$ passes through the y-intercept and the x-intercept of f. Determine the values of a and c. (4)

7.9.6 For which values of x is $f(x) \times h(x) \geq 0$? (2)

FINANCIAL MATHEMATICS

QUESTION 8

8.1 A new fridge costing R22 000 is bought on Hire Purchase. The shop requires a 10% cash deposit and charges an interest rate of 16% p.a.

8.1.1 Determine the value to be paid off on Hire Purchase. (2)

8.1.2 If the shop charges an insurance fee of R30 per month, determine the equal monthly payments if the fridge is paid off over $3\frac{1}{2}$ years. (5)

8.1.3 What is the total amount paid for the fridge? (2)

8.2 Brenda is travelling to Europe and the USA. She wishes to take €750 and \$1000 as she leaves. The exchange rates are : $R1 = 0{,}05427€$ and $\$1 = R17{,}59735$

8.2.1 How many Rand in total does Brenda need to convert to Euros and Dollars? (3)

8.2.2 Based on the exchange rates, determine to the closest \$, the value of €150. (3)

8.3 Sue invested R25 000 in a bank for five years at a compound interest rate of 8% p.a.

8.3.1 How much did Sue have in her account at the end of five years? (3)

8.3.2 Determine the simple interest rate which would have resulted in her having exactly the same amount in her account after five years. (3)

8.4 A car cost R250 000 on 1 January 2024. Assuming an average rate of inflation of 5,5% p.a., calculate, based on inflation alone, the cost of the same car at the end of 2031. Round off your answer to the nearest thousand rand. (3)

8.5 R22 500 was invested at a simple interest rate of 9,5% p.a. for x years. The interest earned amounted to R14 962.50.

8.5.1 Determine the value of x. (3)

8.5.2 If the initial interest rate rather earned compound interest of y% p.a. and led to the same accumulated amount over the same period, determine the value of y. (3)

8.6 Rx is invested and earns compound interest. After 5 years the accumulated amount is R7 308.46. After 10 years this amount grows to R11 244.98.

8.6.1 Determine the interest rate. (5)

8.6.2 Determine the value of x. (2)

8.7 Furniture is advertised at Rx in a store. A 15% deposit is required and thereafter the furniture may be acquired on a Hire Purchase Agreement at 18% p.a. over 3 years. If the equal monthly instalments are R636.32, determine:

8.7.1 the value of x to the nearest rand. (6)

8.7.2 the total cost of the furniture. (2)

8.7.3 the total amount of interest paid. (2)

8.8 Peter and Paula each invest R2000 in a bank for 5 years. Peter's investment earns simple interest at 9% p.a. Paula's investment earns compound interest at 8% p.a.

Using appropriate mathematics, determine which of the two had more in their bank account at the end of the 5 years. (4)

8.9 Julia bought some furniture making use of a Hire Purchase Agreement. The sale price was R50 000. The furniture store required a 7,5% deposit. The Agreement stated that the balance must be paid off in equal monthly payments over 4 years at an interest rate of 17% p.a. A monthly insurance fee of R120 was also to be paid.

8.9.1 Show that the deposit paid by Julia was R3 750.00. (1)

8.9.2 Calculate Julia's equal monthly repayments. (4)

8.10 The average price of petrol in 1973 was approximately 13 cents per litre. The average price of petrol increased to approximately R24.00 per litre in 2023.

Calculate the annual rate of inflation if it is assumed that the rate remained constant over this period of 50 years. (Answer correct to the nearest integer) (3)

8.11 Felix uses Ry for a trip to the USA. The bank first charges a fee of 1,7% on this amount. The balance, being Rx is converted into \$1500 for a trip to the USA. The exchange rate is $R1 = \$0{,}052$.

8.11.1 Determine the value of x. (3)

8.11.2 Hence determine the value of y. (2)

8.11.3 Felix buys an item in a gift shop in the USA which costs €75.30. Effectively how much did he pay for the item in Rand if €1= \$1,08 ? (3)

8.12 A census at the end of 2021 gives the population of South Africa at 59,3 million. The average growth rate over the following 5 years is expected to be 0,9% p.a.

8.12.1 Calculate the expected population of South Africa at the end of 2026, and give your answer to the nearest ten thousand. (3)

8.12.2 If the population at the end of 2031 is actually 66,16 million, determine the actual average growth rate over the 10 years. Answer to two decimal places. (3)

8.13 An item costs R3500 today. Based on inflation alone, determine to the nearest ten rand, the cost of the item 4 years ago if the average rate of inflation was 11% p.a. (3)

8.14 Eli invests R20 000 for five years earning compound interest of 8,5% p.a. After four years he required some funds and he withdrew Rx from his savings account. The accumulated amount at the end of five years was R28 445.63.

Determine the value of x. (5)

8.15 Shelly required R32 000 for a holiday. She invested R15 000 for seven years. The compound interest rate for the first four years was 9% p.a. and thereafter it increased to 10% p.a. After fours years she added R2 000 to her savings account.

8.15.1 Show that Shelly had insufficient funds for her holiday after seven years. (5)

8.15.2 Show that if Shelly left the accumulated amount in her account for a further six months at 10% p.a., she would have sufficient funds for her holiday. (2)

PROBABILITY

QUESTION 9

9.1 Given two events M and N:

$$P(M) = 0{,}30 \quad ; \quad P(N) = 0{,}65 \quad ; \quad P(M \text{ or } N) = 0{,}74$$

9.1.1 Determine the value of $P(M \text{ and } N)$. (3)

9.1.2 State, with reason, if events M and N are mutually exclusive. (2)

9.2 The letters of the word $PROBABILITY$ are placed in a bag. A letter is then randomly removed from the bag. What is the probability that the removed letter:

9.2.1 is not a vowel? (2)

9.2.2 is an A or a B? (2)

9.2.3 is a B ? (1)

9.3 A group of 100 learners were asked if they liked hamburgers or milkshakes. Event H – liked hamburgers. Event M – liked milkshakes.

- 77 liked hamburgers
- 57 liked milkshakes
- x liked both hamburgers and milkshakes
- y liked neither

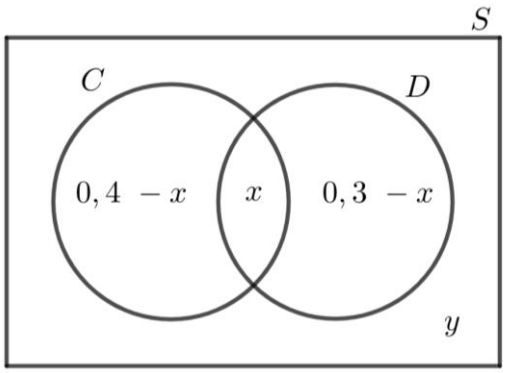
9.3.1 Draw a neat Venn Diagram representing this information. (4)

9.3.2 If $P(H \text{ or } M) = 0{,}92$ determine the value of x. (3)

9.3.3 Hence determine the value of y. (2)

9.3.4 Calculate the probability that a learner chosen at random liked both hamburgers and milkshakes. (Give your answer correct to TWO decimal places) (2)

9.4 Consider the following Venn Diagram, where the full sample space lies within the rectangle. All numbers and variables shown are probabilities.



9.4.1 Write down the values of $P(C)$ and $P(D)$. (2)

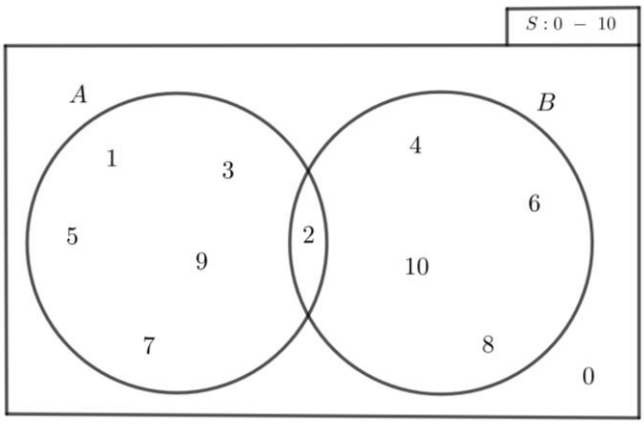
9.4.2 If $P(C \text{ or } D) = 0{,}6$ calculate the value of x and y. (4)

9.4.3 Calculate $P(C' \text{ or } D)$. (3)

9.5 $A = x^2 - x - 12$ for $-4 \leq x < 5$ and $x \in \mathbb{Z}$.

A value for x is chosen randomly within the domain and substituted into the equation. Determine the probability that $A = 0$. (4)

9.6 The first eleven whole numbers are shown in the Venn Diagram below. The events are A and B.



9.6.1 State with reasons if events A and B are: 9.6.1.1 Mutually Exclusive? (2)

9.6.1.2 Complementary? (2) 9.6.1.3 Exhaustive? (2)

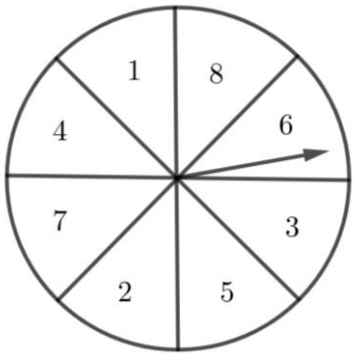
9.6.2 Write down the values of:

9.6.2.1 $n(A)$ (1) 9.6.2.2 $n(A\ and\ B)$ (1) 9.6.2.3 $n(Not\ B)$ (1)

9.6.3 Determine the following probabilities: 9.6.3.1 $P(B)$ (1)

9.6.3.2 $P(Not\ A)$ (1) 9.6.3.3 $P(A\ and\ B)$ (1) 9.6.3.4 $P(Not\ A\ and\ Not\ B)$ (1)

9.7 A spinner with 8 equal segments is shown in the diagram. The arrow is spun once.



Determine the probability of:

9.7.1 obtaining a prime number. (2) 9.7.2 obtaining a multiple of 3. (2)

9.7.3 not obtaining a 1. (1) 9.7.4 obtaining a factor of 8. (2)

9.8 In a school there are 100 Grade 10 pupils.

- 60 take mathematics
- 35 take geography
- 10 take neither mathematics or geography
- x take both mathematics and geography

9.8.1 Draw a neat Venn Diagram representing this data. (4)

9.8.2 How many pupils take both mathematics and geography? (3)

9.8.3 What is the probability that a pupil selected randomly from the grade 10's:

9.8.3.1 takes both mathematics and geography? (1)

9.8.3.2 does not take maths? (1)

9.8.3.3 takes geography but not mathematics? (1)

9.9 Consecutive integers from and including -6 up to and including 6 are written on cards and placed in a bag. A card is randomly drawn from the bag and the integer value written on the card is substituted into the following equation:

$$T = \frac{\sqrt{4-x^2}}{x-1}$$

What is the probability that T will be:

9.9.1 0 ? (2) 9.9.2 real? (3) 9.9.3 undefined? (2)

9.10 100 girls were asked which of two shampoos A or B they liked.

- 35 girls liked shampoo A
- 50 girls liked shampoo B
- 25 girls liked neither shampoo

9.10.1 MAKE USE THE FORMULAE $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ and $P(E) = \frac{n(E)}{n(S)}$ to determine how many girls liked both shampoos A and B. (5)

9.10.2 Let x girls like both shampoos.

9.10.2.1 Draw a neat Venn Diagram representing all this information. (4)

9.10.2.2 Solve for x, and compare your answer to that in 9.10.1 above. (2)

9.10.3 Complete the Venn diagram, then use it to determine the probability that a girl selected at random:

9.10.3.1 liked both shampoos. (1)

9.10.3.2 liked neither shampoo. (1)

9.10.3.3 did not like shampoo B. (2)

9.10.4 Are the events Liking shampoo A and Liking shampoo B mutually exclusive? Explain. (2)

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